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# The Barbelo-Tesla Prime Sieve: A Visual Framework for Prime Distribution via Iterative Digit-Sum Reduction

INTERACTIVE VISUALIZATION · JUNE 2026

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Published at: [insleben.com/Tesla-Sieb/tesla-sieb.html](https://insleben.com/Tesla-Sieb/tesla-sieb.html)

## ABSTRACT

This paper presents the **Barbelo-Tesla Prime Sieve**, a tabular visualization of prime number distribution based on iterative digit-sum reduction. All natural numbers are classified by their digital root (iterative digit sum) into nine classes (1–9). Classes 3, 6, and 9 are structurally excluded: every member of these classes is divisible by 3 and therefore cannot be prime (with the single exception of 3 itself).

This leaves six active classes — digital roots 1, 2, 4, 5, 7, and 8 — in which all primes greater than 3 must appear. When prime numbers are entered into a table organized by these six classes, a recurring structural pattern becomes visible:

6 candidate positions · 1 certain gap (DS 9) · 2 candidate positions · 1 certain gap (DS 3) · repeating

The three structurally empty classes (DS 3, DS 6, DS 9) act as fixed dividers within this repeating 10-position cycle. Beyond these structural gaps, additional empty

positions arise from multiples of primes themselves — the classical sieving process — which are displayed as a secondary overlay.

## METHOD

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The sieve is constructed as follows:

- Compute the iterative digit sum (digital root) of each natural number.
- Arrange numbers in a table where each column corresponds to one digital-root class.
- Columns DS 3, DS 6, DS 9 remain structurally empty (except the prime 3 in column DS 3).
- Mark primes in the remaining six columns; mark composite members of those columns as sieved.

The result is a visual sieve that separates structural impossibility (digit-sum constraint) from empirical sparsity (composite numbers). The 6+1+2+1 rhythm of candidate and gap positions repeats indefinitely and is invariant across all number ranges.

## IMPLEMENTATION

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The visualization is implemented as a pure HTML5 interactive table with no external dependencies. An adjustable range slider controls how many numbers are displayed (up to 1,000). A toggle allows composites to be shown or hidden. Hovering over any cell reveals the full digit-sum chain for that number.

## RELATION TO PRIOR WORK

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This visualization is part of a series of prime number representations by the same author. The *Barbelo Prime Tree I* (DOI: 10.5281/zenodo.20475174) and *Barbelo Prime Tree II* (DOI: 10.5281/zenodo.20497826) organize primes as branching tree structures based on prime gaps and the OEIS sequence A333489 respectively. The Barbelo Prime Sieve approaches distribution from a digit-theoretic perspective, complementing the geometric tree representations.

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Related: [Barbelo Prime Tree I](#) · [Barbelo Prime Tree II](#)